Internship Report

Health Risk Classification for Chronic Diseases

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1. Introduction

Chronic diseases such as diabetes and cardiovascular conditions are often preceded by measurable clinical and lifestyle risk factors, making early identification of at-risk patients critical for timely intervention. This project, undertaken as part of a data analytics and AI internship, focuses on building a Healthcare Risk Classification System that predicts a patient's health risk level using clinical and lifestyle data. By applying machine learning to attributes such as age, BMI, blood pressure, cholesterol, and HbA1c levels, the system supports proactive, data-driven clinical decision-making and demonstrates how predictive analytics can assist healthcare providers in prioritizing patient care.

2. Abstract

The objective of this project is to classify patients into four health risk categories — Low, Medium, High, and Critical — based on a balanced dataset of 1,000 patient records containing 15 clinical and lifestyle features, including Age, BMI, Systolic/Diastolic Blood Pressure, Cholesterol, HbA1c percentage, Smoking Status, Physical Activity, Sleep Hours, Heart Rate, Daily Steps, and family history of diabetes and heart disease. Data preprocessing involved separating the target variable (Risk_Category), label-encoding it, and building a Scikit-learn preprocessing pipeline that applies standard scaling to numeric features and one-hot encoding to categorical features. A Random Forest Classifier (100 estimators, balanced class weights, max depth of 15) was trained on an 80/20 train-test split to predict patient risk level. The model was evaluated using accuracy score, a classification report, and cross-validation, achieving a reported test accuracy of approximately 95%. Model explainability was implemented using SHAP (TreeExplainer) to identify the health factors most influential to each prediction. The trained model was deployed through an interactive Streamlit web application offering a population-level dashboard, patient database search with AI diagnostics, a manual clinical assessment tool, and automated PDF patient risk report generation, providing a practical decision-support tool for healthcare risk screening.

3. Tools Used

Category	Tools / Technologies
Programming	Python
Data Handling	Pandas, NumPy
Machine Learning	Scikit-learn (Random Forest Classifier, Logistic Regression, StandardScaler, OneHotEncoder, Pipeline)
Explainable AI	SHAP (TreeExplainer)
Visualization	Matplotlib, Plotly
Web Application	Streamlit
Model Persistence	Joblib
Report Generation	ReportLab (PDF patient risk reports)
Version Control	Git & GitHub

4. Steps Involved in Building the Project

- Problem identification: defined the need to classify patients into four risk categories (Low, Medium, High, Critical) using clinical and lifestyle indicators.

- Dataset collection: used a balanced healthcare dataset of 1,000 patient records with 15 clinical and lifestyle features and a Patient_ID identifier.
- Data cleaning and preprocessing: removed the non-predictive Patient_ID column, label-encoded the target variable, and applied standard scaling to numeric features and one-hot encoding to categorical features via a Scikit-learn ColumnTransformer pipeline.
- Exploratory Data Analysis: examined risk distribution, age-versus-BMI relationships, and lifestyle risk correlations within the Streamlit dashboard module.
- Model training and testing: trained a Random Forest Classifier (100 estimators, balanced class weights, max depth 15) on an 80/20 stratified train-test split.
- Model evaluation: assessed performance using accuracy score, a classification report, confusion matrix, and cross-validation scores, achieving approximately 95% test accuracy.
- Explainability using SHAP: applied a SHAP TreeExplainer to compute per-patient feature impact scores, highlighting the most influential health factors behind each prediction.
- Streamlit application development: built an interactive web application with a population health dashboard, a patient database search with AI-based diagnostics, a manual clinical assessment tool, and automatic PDF patient risk report generation using ReportLab.
- GitHub documentation and project organization: structured the repository into data, models, and application files, with a detailed README describing features, workflow, and usage instructions.

5. Conclusion

The project successfully demonstrates an end-to-end machine learning pipeline for healthcare risk classification, from data preprocessing and model training to explainable predictions and a deployed interactive application. The Random Forest model achieved strong predictive performance, and SHAP-based explainability provided transparency into the clinical and lifestyle factors driving each risk prediction, an important consideration for healthcare applications. By surfacing high-risk patients early, the system can support clinicians in prioritizing follow-up care and preventive interventions. The project strengthened practical skills in data preprocessing pipelines, classification modeling with Scikit-learn, model explainability with SHAP, and building data-driven web applications with Streamlit. As noted in the project documentation, potential future improvements include integration with real hospital Electronic Health Records (EHR), real-time patient monitoring, cloud deployment for scalability, a mobile-friendly dashboard, and integration with wearable health devices.